ABOLORE SANNI

AI/Agent Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

AI/Agent Engineer and full-stack developer with hands-on experience designing, building, and operating production AI systems. Builds multi-agent orchestration with LangGraph, LangChain, and the Microsoft Azure AI ecosystem (Foundry Agent Service, Copilot Studio, Entra Agent ID, Agent 365), plus retrieval-augmented generation (RAG) pipelines over ChromaDB and Azure AI Search. Built the Talyn AI learning platform end to end: FastAPI microservices, PostgreSQL, JWT authentication with RBAC, signed payment webhooks, AWS S3 storage, and a React and TypeScript frontend on Docker with GitHub Actions CI/CD. Microsoft Certified: Azure AI Fundamentals. A B.Sc. in Psychology underpins human-centred agent design.

TECHNICAL SKILLS

AI and Agent Engineering: LangGraph, LangChain, Anthropic Claude API, OpenAI GPT-4, Gemini API, Hugging Face Transformers, Microsoft Foundry Agent Service, Copilot Studio, Model Context Protocol (MCP), A2A, multi-agent orchestration, prompt engineering

RAG and Knowledge Systems: Retrieval-augmented generation (RAG), ChromaDB, Azure AI Search, LangChain Retrievers, document chunking, content grounding, vector search, semantic ranking

Languages: Python, JavaScript, TypeScript, SQL, Bash, HTML, CSS

Backend and APIs: FastAPI, REST API design, API versioning, Pydantic, SQLAlchemy ORM, Alembic, Uvicorn, JWT authentication, refresh token rotation, RBAC, webhooks, rate limiting, pytest

Frontend: React, React Router, Zustand, Tailwind CSS, Vite, ESLint, Vitest, Testing Library

Machine Learning: PyTorch, scikit-learn, DistilRoBERTa, sentiment and emotion classification, data annotation, automated evaluation, golden dataset testing

Databases, Infrastructure and Cloud: PostgreSQL, MySQL, Supabase (Auth, Row-Level Security), Redis, Docker, Docker Compose, Caddy, Git, GitHub Actions, CI/CD, Microsoft Azure, Azure Container Registry, AWS S3

Integrations, Security and Governance: Paystack, SerpAPI, eBay API, Google Shopping, pypdf, SMTP, Row-Level Security, secret management, ClamAV malware scanning, content filtering, agent observability, agent identity, structured logging

PROFESSIONAL EXPERIENCE

AI/Full-Stack Developer (Contract) | Talyn - AI Learning Engagement Platform
2026 - Present | Lagos, Nigeria

- Built Talyn, an AI-powered learning engagement platform, working from product research (learner questionnaires and behavioural-science work on course dropout) that shaped the roadmap for individual, institutional, and corporate learning markets.
- Delivered across the full stack as the developer: system architecture, backend and AI services, frontend, deployment, and testing, from requirements to production on a three-service platform (FastAPI backend, React and TypeScript frontend, Claude-powered AI coach microservice) serving distinct learner, creator, and admin roles.
- Established CI/CD on GitHub Actions with a real PostgreSQL service container, automated pytest and Vitest suites, Docker image builds, and API documentation checks gating every merge, plus an operations runbook for backups, restores, and secret rotation.

AI/Agent Engineer | Reliance Infosystems
Jun 2026 - Aug 2026 | Lagos, Nigeria

- **Agent Design and Orchestration:** Design, build, and orchestrate intelligent agents using Microsoft Foundry Agent Service and Copilot Studio extensibility, turning prototypes into production-ready AI solutions.
- **RAG Implementation:** Build retrieval-augmented generation (RAG) and content grounding pipelines that deliver accurate, context-aware agent responses sourced from verified knowledge bases.

- **Engineering Best Practices:** Apply Git version control, automated testing, and CI/CD pipelines with GitHub Actions, GitHub Copilot, VS Code, and Cursor; build modular reusable agent components and establish agent identity and end-to-end observability with Entra Agent ID and Agent 365, meeting architectural, security, and responsible AI governance standards.

AI Developer Intern | Publica AI
Aug 2025 - May 2026 | Lagos, Nigeria

- **End-to-End System Design:** Architected full-stack AI applications from build phase to production using FastAPI, React, and Supabase across the full software development lifecycle, containerising backend and frontend with Docker Compose for reproducible development, staging, and production environments.
- **Hybrid AI Pipelines:** Engineered emotion-detection pipelines fusing local Hugging Face models with cloud LLMs (Gemini, OpenAI) for nuanced multi-dimensional sentiment classification.
- **Agentic Frameworks:** Designed stateful multi-agent document-processing systems with LangGraph, enabling autonomous orchestration of complex research workflows.

Data Annotator | Awarri
Nov 2025 - Jan 2026 | Lagos, Nigeria

- Delivered high-precision annotation of text, image, and audio datasets for large-scale ML training pipelines, maintaining 98% accuracy across 10,000+ labelled data points for Computer Vision and NLP training.

KEY PROJECTS

Talyn - AI Learning Engagement Platform | AI/Full-Stack Developer | 2026

- Architected a FastAPI and PostgreSQL backend across 15 API route modules with JWT access and refresh token rotation, role-based access control (learner, creator, admin) with an immutable audit log, and Google ID-token sign-in.
- Built a Claude (Anthropic) AI coach with 17 endpoints that consume a live LearnerContext (XP, streaks, quiz history, badges, missions, and course progress) loaded server-side over a verified JWT, producing personalised explanations, study plans, quizzes, and learning paths.
- Engineered prompt-level and contract-level testing (76 offline tests against a fake LLM client) so AI regressions are caught in CI with no network access and no API spend, and integrated Paystack payments with HMAC-SHA512 signed webhooks and a single idempotent settlement path, so webhook and redirect races settle exactly once after server-side amount verification.
- Hardened file delivery with S3 presigned POST policies that enforce content-length caps, Content-Type plus extension validation, optional ClamAV malware scanning, and server-side encryption, plus rate limiting, JSON logging with X-Request-ID, and rehearsed pg_dump and pg_restore backups.

Tech Stack: Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Redis, PyJWT, Anthropic Claude API, React, TypeScript, Vite, Zustand, Tailwind CSS, Vitest, Testing Library, pytest, Docker, Docker Compose, Caddy, Paystack, AWS S3, GitHub Actions

ForexGPT - AI-Powered Forex Trading Platform | Full-Stack Developer (Team of 8) | 3-Week Agile Sprint

- Built the FastAPI backend as one of 4 Full-Stack Developers on an 8-person team (alongside 4 AI Engineers) over a 3-week agile sprint.
- Implemented authentication middleware and the routes and services for 5 modules: signal extraction, AI mentor, code generation, backtesting, and quant finance.
- Integrated the backend with a fine-tuned Mistral 7B model via the Hugging Face Inference API and a Supabase (PostgreSQL, RLS) database.
- Collaborated using individual Git branches merged into master via pull requests with team code review.

Tech Stack: Python, FastAPI, Supabase, PostgreSQL, LangChain, Google Gemini, Mistral 7B, Hugging Face Inference API, Git

SoulSync - Hybrid Psychotherapy Wellness Agent | Full-Stack AI Engineer

- Engineered a hybrid AI system pairing local NLP emotion classification (DistilRoBERTa) with generative empathy (Gemini, OpenAI) to deliver secure, context-aware, CBT-informed wellness interventions.
- Built a multi-layered sentiment pipeline where emotion classification informs the LLM prompt context, keeping outputs grounded in structured therapeutic logic rather than open-ended generation.

- Designed a Supabase schema with Row-Level Security (RLS) for full user-data isolation and restructured a monolithic script into a containerised, microservice-ready architecture (FastAPI, React, PostgreSQL) with Docker Compose.

Tech Stack: React, Tailwind CSS, FastAPI, PyTorch, Hugging Face Transformers, Gemini API, LangChain, Supabase, PostgreSQL, Docker, Docker Compose

My-TaxPal - AI Tax Compliance and Advisory Assistant | RAG Engineer

- Built a RAG-powered chatbot demystifying Nigeria's 2025 Tax Reform Bills for 200M+ citizens, indexing official legislative documents in ChromaDB and wiring LangChain Retrievers to ground every answer in verifiable source excerpts, eliminating hallucinations on high-stakes compliance questions.
- Deployed a LangGraph conversation flow for multi-turn advisory sessions, preserving context across complex tax scenarios in plain English.

Tech Stack: React, Python, FastAPI, LangChain, LangGraph, ChromaDB

EDUCATION

B.Sc. Psychology | Lead City University
2019 - 2023 | Ibadan, Nigeria

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals, Microsoft, October 2026
- Generative AI Fellowship - AI Developer Track, Certificate of Completion
- Mastercard Foundation Associate Programme (in partnership with Jobberman) - Prep2Soar Course, Certificate of Achievement, Jobberman Nigeria. Modules: Self Awareness, Growth Mindset, Emotional Intelligence, Work Ethics and Professionalism, Presentation Skills, Business Communication, Problem-Solving and Critical Thinking, Digital Safety and Cybersecurity, Personal Effectiveness and Time Management, and Interview Skills.

LANGUAGES

English (Native), Yoruba (Native)